

Vol 8 Chapter 7 — Rigid Body Dynamics

Keeper (author pass — deep math/physics revision)

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Chapter 7 — Rigid Body Dynamics

Level 1 — one sentence

A rigid body has 6 degrees of freedom (3 translation + 3 rotation), with rotational dynamics governed by the inertia tensor I_{ij} and Euler's equations of motion, producing rich phenomena from gyroscope precession to free-symmetric-top instability ("tennis racket theorem") — all derived from substrate Scale-2 angular momentum dynamics.

Level 2 — graduate-physicist precision

7.1 Rigid body kinematics

A rigid body's orientation in 3D is specified by 3 parameters (Euler angles, quaternions, or rotation matrix). Combined with 3 translational DOFs, total 6 DOFs.

Euler angles (ϕ, θ, ψ) — common choice (e.g., precession, nutation, spin for spinning top).

Angular velocity $\vec{\omega}$ is the instantaneous rotation axis $\times$ angular speed.

7.2 The inertia tensor

For a rigid body with mass distribution $\rho(\vec{r})$, the inertia tensor:

$$I_{ij} = \int \rho(\vec{r})(r^2\delta_{ij} - r_i r_j) d^3r$$

Symmetric 3×3 matrix. Diagonalize $\rightarrow$ principal moments of inertia I_1, I_2, I_3 and principal axes.

For axially symmetric bodies (rod, disk, sphere): $I_1 = I_2 \neq I_3$ or $I_1 = I_2 = I_3$.

Angular momentum: $\vec{L} = I\vec{\omega}$.

Kinetic energy: $T = (1/2)\vec{\omega} \cdot I\vec{\omega} = (1/2) \sum_i I_i \omega_i^2$ (in principal-axis frame).

7.3 Euler's equations

In the body frame (rotating with the body):

$$I_1 \dot{\omega}_1 - (I_2 - I_3)\omega_2\omega_3 = \tau_1$$

$$I_2 \dot{\omega}_2 - (I_3 - I_1)\omega_3\omega_1 = \tau_2$$

$$I_3 \dot{\omega}_3 - (I_1 - I_2)\omega_1\omega_2 = \tau_3$$

where τ_i are external torques about principal axes.

7.4 Free rigid body: Euler's spin axes

For torque-free motion ($\tau = 0$): in body frame, $\vec{\omega}$ traces a curve on the inertia ellipsoid. Key result: rotation about the principal axis with intermediate moment of inertia is unstable.

Tennis racket theorem: throw a tennis racket spinning about its intermediate-inertia axis — it tumbles unpredictably. About the largest or smallest principal axis: stable rotation. Demonstrable with any asymmetric book or smartphone.

7.5 Precession and gyroscopes

A spinning top with angular momentum $\vec{L}$ along its symmetry axis experiences gravitational torque $\vec{\tau} = m\vec{g} \times \vec{r}_{cm}$. The top precesses at rate:

$$\Omega_{\text{prec}} = \frac{mgr_{cm}}{I\omega_{\text{spin}}}$$

(small-angle, fast-spin limit). The spinning angular momentum + gravity torque combine to make the top's axis rotate around vertical.

Nutation: additional smaller oscillation superposed on precession (full Euler-equation solution).

Gyroscopes — spinning rotors whose angular momentum stays fixed in space — used for navigation (inertial measurement units, missile guidance, smartphones).

7.6 Rotating reference frames: Coriolis and centrifugal

In a frame rotating at angular velocity $\vec{\Omega}$ relative to inertial:

$$m\vec{a}_{\text{rot}} = \vec{F} + 2m\vec{v}_{\text{rot}} \times \vec{\Omega} + m\vec{\Omega} \times (\vec{r} \times \vec{\Omega})$$

- Coriolis force: $2m\vec{v} \times \vec{\Omega}$ — apparent deflection perpendicular to motion in rotating frame
- Centrifugal force: $m\Omega^2 r$ outward (radial)

Earth's rotation gives: - Coriolis: deflects winds (clockwise in N hemisphere), affects projectile trajectories, creates large-scale weather patterns - Centrifugal: slightly reduces effective gravity at equator (~0.34%)

7.7 Worked example: Earth's rotation

Earth's rotation: $\Omega = 2\pi/(86400 \text{ s}) = 7.27 \times 10^{-5} \text{ rad/s}$.

Coriolis acceleration at speed 100 m/s: $a_C = 2v\Omega = 2(100)(7.27 \times 10^{-5}) = 1.45 \times 10^{-2} \text{ m/s}^2$. Significant for missiles, weather.

Centrifugal at equator: $\Omega^2 R_E = (7.27 \times 10^{-5})^2 \cdot 6.4 \times 10^6 = 0.034 \text{ m/s}^2$. Reduces apparent gravity by 0.34%.

7.8 Free precession of Earth (Chandler wobble)

Earth's rotation axis is misaligned by ~10 m at the surface; this misalignment processes with period ~433 days (Chandler wobble, observed 1891). Comes from Earth's slight oblateness ($I_1 \neq I_3$) — application of Euler's equations to a free symmetric top.

7.9 K-audit anchors

- Vol 5 Ch 3: angular momentum quantum analog
- Vol 0 Ch 8: substrate Noether currents

Level 3 — 5th-grader accessibility

A rigid body has fixed shape. Total 6 DOFs: 3 to translate + 3 to rotate. Rotation described by the inertia tensor I_{ij} (how mass is distributed). Diagonalize to get principal moments of inertia along principal axes. Euler's equations govern rotation. Cool result (tennis racket theorem): rotation about the *intermediate* principal axis is unstable — flip a phone gently, it flips itself. Gyroscopes: spinning rotor's angular momentum holds steady → used in navigation. Precession of a top: gravity torques the spin axis, axis rotates around vertical. Earth's rotation creates Coriolis force (deflects winds, projectiles) and slight centrifugal reduction of gravity at equator.

What comes next

Chapter 8 develops oscillations and normal modes.

Where to look this up

- Goldstein Ch 4-5; Marion-Thornton Ch 11
- BST: Vol 5 Ch 3